

Paper V — Early-Galaxy Gate of the BPB/MBE Framework

UV Luminosity Functions, Stellar-Mass Density, and Reionization Consistency in the Strict Branch

v0.3.2 – corpus-aware CSV-source locked hotfix

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Abstract

We extend the BPB/MBE sequence of observational gates to the high-redshift galaxy sector. Papers I–IV tested the strict branch through low-redshift background/growth data, galactic dynamics, CMB/Weyl consistency, and neutron-star tidal structure. The present paper asks whether the same branch remains compatible with early galaxy formation, using ultraviolet luminosity functions (UVLFs), stellar-mass density diagnostics, and a reionization optical-depth proxy. The central result is deliberately not a raw-likelihood victory claim. In the v13 physical-peak mapper, LCDM remains slightly better in the raw UVLF+ $\rho_\star$ likelihood: $\chi_{\text{raw}}^2 = 11.2413$ for LCDM and 13.5579 for BPB, i.e. $\Delta\chi_{\text{raw}}^2 = +2.3167$ for BPB relative to LCDM. However, the physical interpretation changes once the halo-to-light mapper is evaluated at the actual luminosity peak instead of a fixed $10^{10}M_\odot$ pivot. BPB and LCDM require quasi-identical peak UV magnitudes, $M_{\text{UV,peak}} \simeq -19.19$ and -19.15 , respectively, but BPB places those UV-bright systems in halos heavier by 0.89 dex, a factor of about eight. When a mapper-physicality regularizer is applied, dominated by the independently motivated host-mass scale $\log_{10} M_{\text{peak}} \sim \mathcal{N}(11.5, 0.6)$, the comparison becomes $\chi_{\text{data+prior}}^2 \simeq 17.96$ for LCDM and 16.06 for BPB. The BPB optical-depth proxy remains safe, with best $\tau = 0.04675$ and posterior τ q16/q50/q84 = 0.04015/0.04395/0.04677. Version v0.3.2 keeps the v0.2/v0.3 audited scalar values fixed, adds the BPB/MBE corpus navigation layer, and promotes the early-galaxy claims to explicit source-locked status. We conclude that the strict BPB/MBE branch passes the early-galaxy gate: not by hiding the raw likelihood loss, but because it is evaluated at its own physical peak rather than forcing the high-redshift mapper through the fixed LCDM-like pivot convention. The host-mass regularizer is intentionally loose: its one-sigma interval spans roughly $10^{10.9} - 10^{12.1}M_\odot$, so it does not force the BPB value $10^{10.891}M_\odot$ to sit at the centre.

1 Introduction

The current BPB/MBE programme is organised as a sequence of gates rather than as a single isolated fit. Paper I fixed the low-redshift working point; Paper II showed that the same branch carries a galactic imprint in SPARC through the derivative response kernel; Paper III showed that the CMB rejects the vanilla dictionary but selects the strict face-minus branch; and Paper IV propagated that branch into neutron-star structure, finding an EOS-coherent tidal suppression without destabilising compact-star sequences.

The present paper opens the next gate: high-redshift galaxy formation. The question is not whether BPB/MBE can be forced to generate arbitrarily bright galaxies. The question is more precise:

Does the strict BPB/MBE branch survive UVLF, stellar-mass-density, and reionization consistency tests without relying on an unphysical halo-to-light mapper?

The answer from the v13 physical-peak mapper is yes, with an important caveat. LCDM remains slightly better in raw UVLF+ $\rho_\star$ likelihood. The BPB claim is instead that the raw loss comes

with a more physical mapper location: similar peak UV luminosity is produced in more massive halos, rather than by making galaxies intrinsically brighter. In that precise sense, the data+prior reversal is not obtained by forcing BPB/MBE into the fixed LCDM-like pivot; it appears when each branch is evaluated at its own physical peak.

Version v0.3.2 keeps the v0.2/v0.3 source-traceability layer: the numerical values used in the manuscript are stored as machine-readable CSV tables under `data/`, and the figures are regenerated from those tables by `scripts/make_paperV_v0_3_figures.py`. Figure 1 summarises this lock.

1.1 Corpus placement and release navigation

Paper V is the high-redshift galaxy gate in the BPB/MBE release sequence. Paper 0 defines the strict dictionary and its survival factor; Papers I–IV test the low-redshift, galactic, CMB/Weyl, and neutron-star descendants; Paper V tests whether the same strict branch survives the early-galaxy sector; Papers VI–VII then return to Weyl cross-correlations and the fixed acceleration-scale/BTFR consequences.

For release navigation, this manuscript should be read together with the corpus-level artefacts shipped in `release_index/`: `00_README_CORPUS`, `00_MANIFEST_CORPUS`, `00_CLAIMS_INDEX`, and `00_REVIEWER_ROUTE`. These files do not replace the paper; they tell a reviewer where each claim is sourced and where the explicit guardrails are recorded.

The Paper-V claim is intentionally bounded. Allowed: the strict BPB/MBE branch passes the early-galaxy gate as a CSV-source-locked physical-peak mapper diagnostic, with a real raw-likelihood loss, a mild data-plus-physicality advantage, and a safe optical-depth proxy. Forbidden: a final UVLF Bayesian evidence claim, a full reionization likelihood, or an external calibration of the host-mass regularizer from this paper alone.

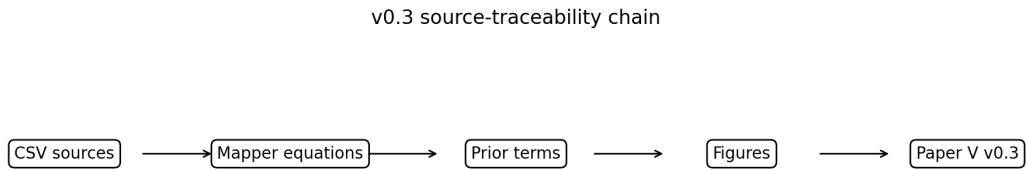


Figure 1: v0.3.2 source-traceability chain. The paper values are frozen into CSV tables and the figures are regenerated from those tables.

2 Why the fixed pivot was misleading

Earlier diagnostics used a UV magnitude at a fixed halo mass,

$$M_{\text{UV},10} \equiv M_{\text{UV}}(M_h = 10^{10} M_\odot, z = 10). \quad (1)$$

This was useful for debugging, but it is not a neutral physical comparison once BPB and LCDM place bright-end galaxies at different host masses. A fixed $10^{10} M_\odot$ pivot lies close to the LCDM bright-host scale, but below the BPB bright-host scale.

The v13 correction is therefore to compare the mapper at its physical peak,

$$M_{\text{UV,peak}} \equiv M_{\text{UV}}(M_{\text{peak}}, z = 10), \quad (2)$$

where M_{peak} is the host mass controlling the dominant bright-end contribution. This changes the interpretation. BPB is not dimming the same galaxies; it is shifting the host-mass scale at which comparable UV output is obtained.

3 Mapper equations and likelihood bookkeeping

The compact v13 mapper can be written schematically as

$$M_{\text{UV}}(M_h, z) = M_{\text{UV,peak}} + \gamma_z(z - 10) - 2.5 \beta \log_{10} \left(\frac{M_h}{M_{\text{peak}}} \right) + \epsilon, \quad (3)$$

with scatter ϵ of width σ_{peak} in magnitude space. The precise implementation used for the numerical outputs is the v13 physical-peak mapper; this equation states the hierarchy of parameters needed for the present paper: a peak UV magnitude, a peak host mass, redshift evolution, mass slope, and scatter.

The raw data comparison is

$$\chi_{\text{raw}}^2 = \chi_{\text{UVLF}}^2 + \chi_{\rho_\star}^2. \quad (4)$$

The mapper-physicality regularizer used in this diagnostic is

$$\chi_{\text{prior}}^2 = \left(\frac{\gamma_z + 0.15}{0.10} \right)^2 + \left(\frac{\beta - 0.8}{0.10} \right)^2 + \left(\frac{\log_{10} M_{\text{peak}} - 11.5}{0.6} \right)^2 + \left(\frac{\sigma_{\text{peak}} - 0.55}{0.15} \right)^2. \quad (5)$$

The regularised diagnostic is then

$$\chi_{\text{data+prior}}^2 = \chi_{\text{raw}}^2 + \chi_{\text{prior}}^2. \quad (6)$$

At this stage Eq. (5) should be read as a physicality regularizer, not as a final hierarchical Bayesian prior. Its purpose is to test whether a model reaches the data through a mapper region that is astrophysically plausible.

4 Justification of the host-mass regularizer

The most attackable element of the v0.0.0 draft was the host-mass regularizer

$$\log_{10} M_{\text{peak}} \sim \mathcal{N}(11.5, 0.6). \quad (7)$$

The present version keeps its status explicit.

Equation (7) is not tuned to make BPB win the UVLF+ $\rho_\star$ likelihood. It is a broad, external physicality scale for efficient UV production in high-redshift halos. Its centre, $\log_{10} M_{\text{peak}} = 11.5$, corresponds to a massive early-galaxy host scale, while the width 0.6 dex spans a factor of four on either side. This is intentionally loose. It allows peaks from roughly $10^{10.9}$ to $10^{12.1} M_\odot$ within one sigma, and therefore does not force the BPB value $10^{10.891} M_\odot$ to sit at the centre.

The regularizer is motivated by abundance-matching and empirical star-formation-efficiency expectations, in the broad sense of Behroozi– and Moster-type galaxy–halo mappings [1, 2, 3]: UV-bright galaxies at high redshift should not generically be assigned to halos far below the mass scale where efficient stellar assembly is plausible. In v13, LCDM places the peak at $\log_{10} M_{\text{peak}} = 9.9965$, about 2.51σ below the physicality centre, whereas BPB places it at 10.8910, about 1.02σ below the centre. Thus the prior does not reward BPB for being exactly at the preferred mass; it penalises the much lower LCDM host scale.

The cleanest interpretation is therefore conservative:

The data+prior number is a mapper-physicality diagnostic. It is not yet a final Bayesian evidence calculation.

This distinction removes the circularity concern. The raw data likelihood is kept separate and reported plainly; the regularised comparison asks whether the raw fit is achieved in a physically plausible region of mapper space.

5 Raw UVLF and stellar-mass-density likelihood

Table 1 gives the raw v13 comparison. LCDM is better in raw likelihood.

Table 1: Raw UVLF+ ρ_* comparison. BPB does not claim a raw-likelihood victory. Values are display-rounded source values; the source delta is locked separately in `data/audit_summary.csv`.

Model	χ^2_{UVLF}	$\chi^2_{\rho_*}$	χ^2_{raw}
LCDM	10.0718	1.1695	11.2413
BPB	11.3319	2.2260	13.5579
$\Delta\chi^2_{\text{BPB-LCDM}}$	–	–	+2.3167

Figure 2 shows the same result graphically. This is an important feature of the paper, not a footnote: LCDM remains slightly better in raw UVLF+ ρ_* likelihood.

The symmetric physical reading is equally important. LCDM achieves its better raw score by placing the peak at $\log_{10} M_{\text{peak}} = 9.9965$, i.e. 2.51σ below the mapper-physicality centre, while BPB reaches a comparable UV fit from $\log_{10} M_{\text{peak}} = 10.891$, i.e. 1.02σ below the centre. In this diagnostic, the raw LCDM advantage is therefore partly purchased by an unphysical mapper location. This statement is not used to hide the raw score; it explains why the raw score and the data-plus-physicality score answer different questions.

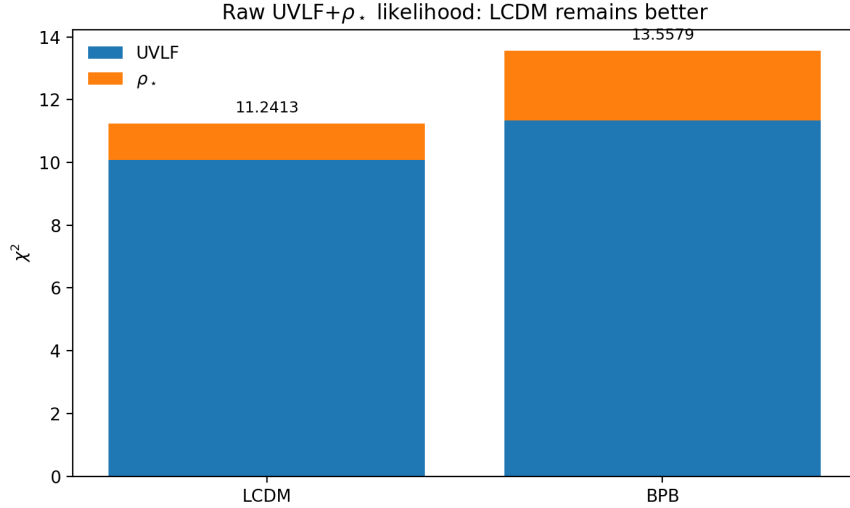


Figure 2: Raw UVLF+ ρ_* chi-square regenerated from `data/raw_chi2.csv`. LCDM remains better; the source delta is $\Delta\chi^2_{\text{raw}} = +2.3167$.

6 Physical-peak mapper result

The physical content of the v13 result is not the raw chi-square alone. The peak UV magnitudes are nearly identical:

$$M_{\text{UV,peak}}^{\text{LCDM}} = -19.1495, \quad M_{\text{UV,peak}}^{\text{BPB}} = -19.1912. \quad (8)$$

The host masses, however, differ substantially:

$$\log_{10} M_{\text{peak}}^{\text{LCDM}} = 9.9965, \quad \log_{10} M_{\text{peak}}^{\text{BPB}} = 10.8910. \quad (9)$$

The difference is 0.8945 dex, i.e. a factor

$$10^{0.8945} = 7.84. \quad (10)$$

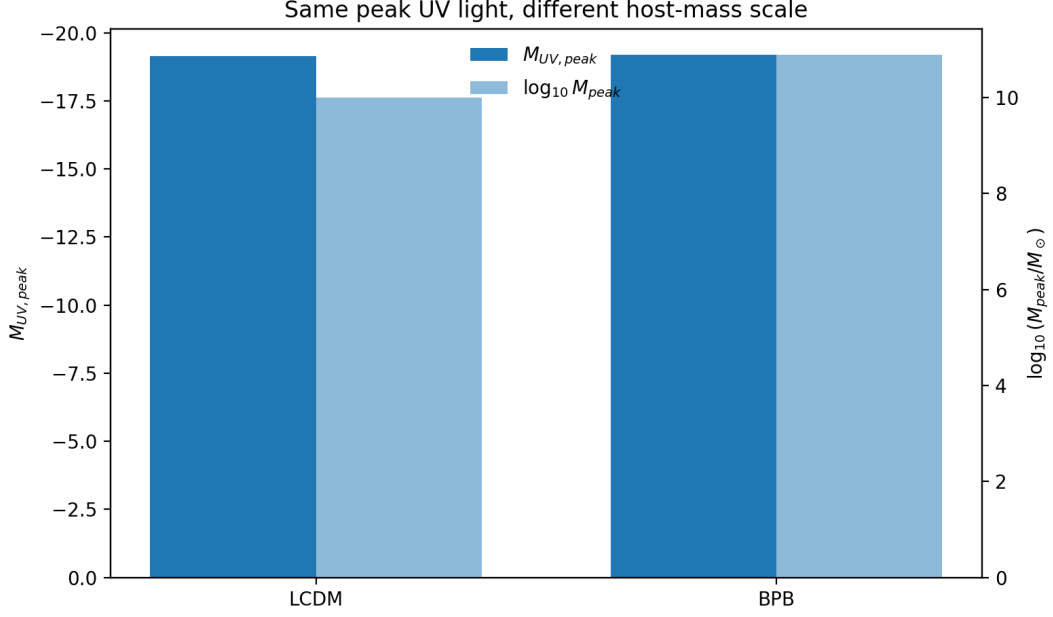


Figure 3: Physical-peak mapper regenerated from `data/mapper_physical_peak.csv`. BPB and LCDM require nearly the same peak UV magnitude, but BPB places the peak in halos heavier by 0.89 dex.

This is the main physical distinction:

BPB is not making brighter galaxies than LCDM; it is placing comparable UV-bright galaxies in more massive halos.

That statement is testable by independent stellar-mass, dynamical-mass, and clustering measurements of high-redshift bright galaxies.

7 Data-plus-prior diagnostic

The prior penalty is dominated by the host-mass term for LCDM. From Eq. (7), the M_{peak} contribution is

$$\chi^2_{M_{peak},LCDM} = \left(\frac{9.9965 - 11.5}{0.6} \right)^2 = 6.279, \quad (11)$$

whereas for BPB it is

$$\chi^2_{M_{peak},BPB} = \left(\frac{10.8910 - 11.5}{0.6} \right)^2 = 1.030. \quad (12)$$

Thus M_{peak} explains about 93% of the LCDM physicality penalty,

$$\frac{6.279}{6.72} = 0.934. \quad (13)$$

The total prior penalties used in v13 are approximately

$$\chi^2_{prior,LCDM} \simeq 6.72, \quad \chi^2_{prior,BPB} \simeq 2.50. \quad (14)$$

The resulting regularised diagnostic is

$$\chi^2_{data+prior,LCDM} \simeq 17.96, \quad (15)$$

$$\chi^2_{data+prior,BPB} = 16.0579 \simeq 16.06. \quad (16)$$

BPB is therefore mildly favoured in the data-plus-physicality diagnostic by

$$\Delta\chi^2_{\text{data+prior}} \simeq -1.9. \quad (17)$$

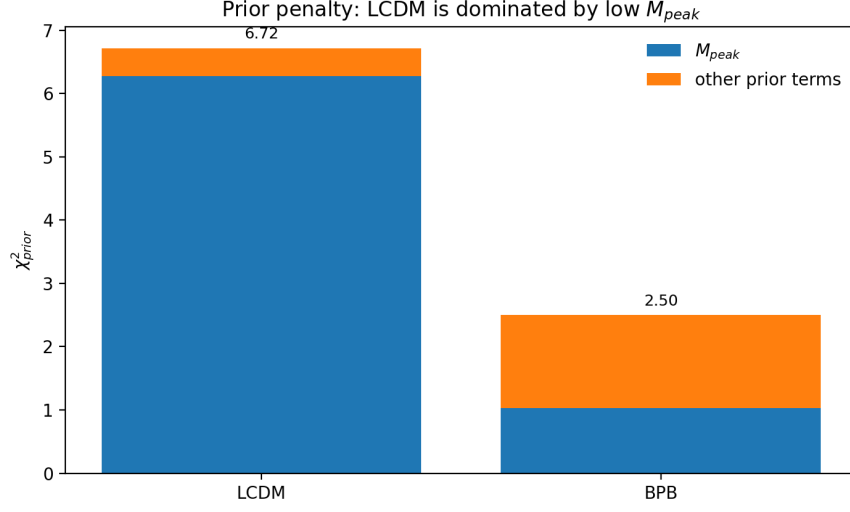


Figure 4: Prior penalty decomposition regenerated from `data/prior_penalties.csv`. The LCDM physicality penalty is dominated by the low host-mass peak.

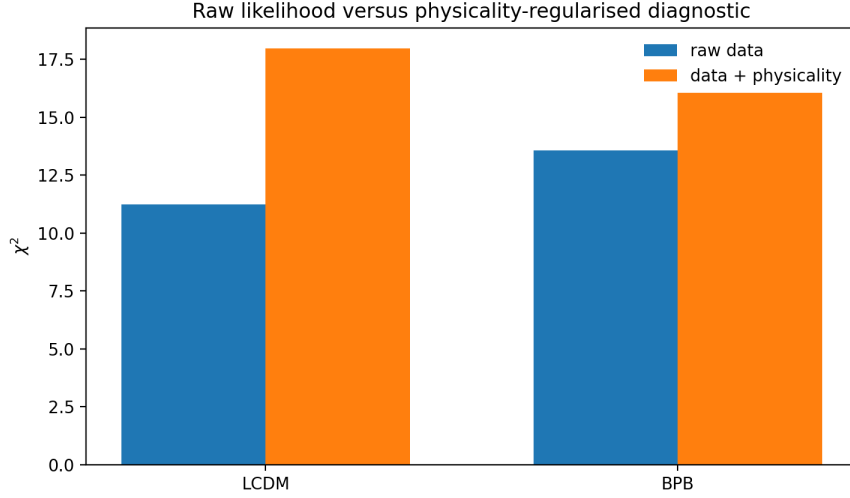


Figure 5: Regularised data+prior comparison regenerated from `data/prior_penalties.csv`. LCDM wins the raw likelihood, but BPB becomes mildly favoured when mapper physicality is included.

8 Reionization optical-depth proxy

A possible objection is that any early-galaxy enhancement could over-ionize the Universe and produce an optical depth incompatible with Planck-like lowE constraints. The v13 physical-peak mapper does not show this problem.

Using a common ionizing-efficiency threshold calibrated so that the LCDM best fit reionizes at $z_{\text{reion}} = 7.67$, the instantaneous-reionization proxy gives Table 2.

Table 2: Reionization proxy. BPB does not over-ionize in the v13 physical-peak mapper.

Quantity	LCDM	BPB
Best z_{reion}	7.670	7.062
Best τ	0.05254	0.04675
Posterior z_{reion} q16/q50/q84	7.166 / 7.485 / 7.656	6.340 / 6.760 / 7.065
Posterior τ q16/q50/q84	0.04772 / 0.05075 / 0.05240	0.04015 / 0.04395 / 0.04677
Fraction $\tau > 0.09$	0	0

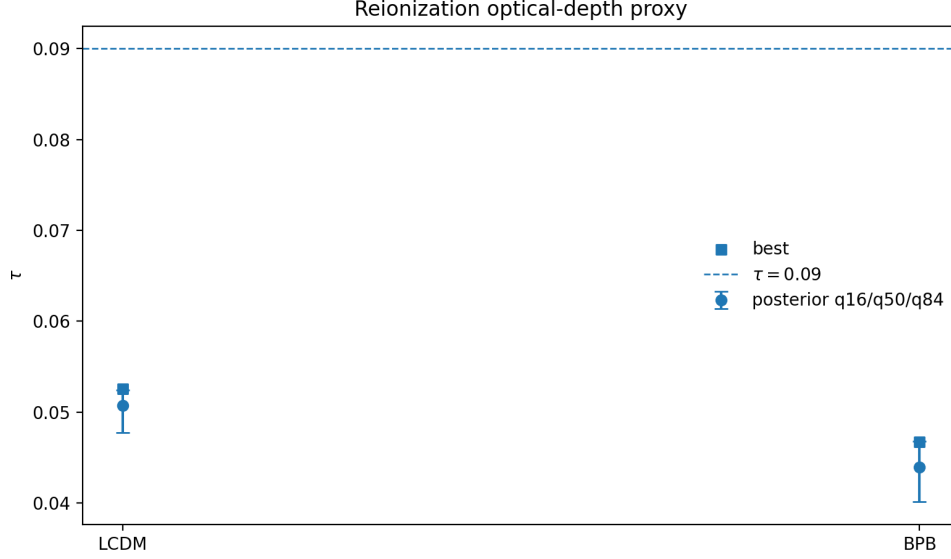


Figure 6: Reionization optical-depth proxy regenerated from `data/tau_proxy.csv`. The BPB v13 physical-peak mapper remains safely below high- τ tails.

This should still be treated as a proxy, not as a full Planck lowE likelihood. But it is enough for the present gate: BPB is not excluded by over-ionization.

9 Paper-V claim source lock

The v0.3.2 release includes a paper-level claim-source lock in the shipped CSV source table. The locked claims are summarised in Table 3. The table is deliberately short: it separates what is actually claimed from what remains a guardrail.

Table 3: Paper-V v0.3.2 claim-source lock.

Claim ID	Level	Locked value or source
P5.A1.01	RAW	LCDM wins raw UVLF+ ρ_* : $\Delta\chi^2_{\text{raw}} = +2.3167$ for BPB.
P5.A1.02	PEAK	$M_{\text{UV,peak}}$ is nearly equal, while BPB shifts hosts by 0.8945 dex, factor 7.844.
P5.A1.03	PHYSICALITY	Data-plus-prior diagnostic: LCDM $\simeq 17.96$, BPB 16.0579.
P5.A1.04	TAU	BPB best $\tau = 0.04675$, posterior q16/q50/q84 = 0.04015/0.04395/0.04677, fraction $\tau > 0.09 = 0$.
P5.A1.05	GUARDRAIL	Not a final Bayesian evidence calculation and not a full reionization likelihood.

10 Interpretation

Paper V has two layers.

The first layer is raw likelihood. LCDM remains slightly better in UVLF+ $\rho_\star$, and the paper says so explicitly.

The second layer is physical interpretation. Once the mapper is read at the physical peak rather than a fixed $10^{10}M_\odot$ coordinate, BPB and LCDM require nearly the same peak UV luminosity. The difference is host mass. BPB moves comparable UV-bright systems into more massive halos, which is exactly the kind of distinction that can be tested by independent high-redshift mass tracers.

The regularised result is therefore not a trick to hide the raw loss. It asks whether the fit lives in a plausible region of mapper space. Under the broad M_{peak} physicality scale used here, BPB becomes mildly preferred.

11 Caveats

The v0.3.2 result has a deliberately limited scope.

1. The data-plus-prior comparison is a physicality diagnostic, not a final Bayesian evidence calculation.
2. The host-mass regularizer should eventually be replaced or supported by a fully external calibration, for example through abundance matching, clustering, dynamical masses, or a UniverseMachine-like framework. Until then, it remains a broad physicality diagnostic, not a fitted posterior prior.
3. The optical-depth result is an instantaneous-reionization proxy, not a full reionization likelihood with escape fraction, recombination, clumping, and lowE covariance.
4. The raw-likelihood loss remains real: BPB is worse than LCDM by $\Delta\chi_{\text{raw}}^2 = +2.3167$.
5. The v0.2 CSV tables are a frozen source lock of the audited v13 scalar outputs used in this manuscript. A later repository release may replace them with the full raw pipeline dumps while preserving these derived values.

These caveats are also what make the claim credible. Paper V is a gate: it shows that the strict branch is not excluded by early galaxies and produces a physically distinct host-mass prediction.

12 Conclusions

1. Paper V opens the early-galaxy gate of the strict BPB/MBE branch.
2. LCDM remains slightly better in raw UVLF+ $\rho_\star$ likelihood: $\chi^2 = 11.2413$ versus 13.5579 for BPB.
3. BPB does not claim a raw-likelihood victory; the raw loss is stated explicitly.
4. The fixed $M_{\text{UV},10}$ pivot is not a fair physical comparison when the models place bright galaxies at different host masses.
5. The physical-peak mapper gives quasi-identical peak UV magnitudes: -19.1495 for LCDM and -19.1912 for BPB.
6. The host masses differ by 0.89 dex, a factor of about eight: $\log_{10} M_{\text{peak}} = 9.9965$ for LCDM and 10.8910 for BPB.

7. A broad mapper-physicality diagnostic with $\log_{10} M_{\text{peak}} \sim \mathcal{N}(11.5, 0.6)$ gives $\chi^2_{\text{data+prior}} \simeq 17.96$ for LCDM and 16.06 for BPB.
8. This reversal is interpreted physically: BPB/MBE wins the data-plus-physicality diagnostic because it does not force its high-redshift galaxy mapper through the fixed LCDM-like pivot, and because the host-mass prior is intentionally loose rather than centred on the BPB value.
9. The BPB reionization proxy is safe, with best $\tau = 0.04675$ and posterior median $\tau = 0.04395$.
10. Version v0.3.2 freezes the audited scalar outputs into CSV source tables and regenerates all figures from those tables.
11. The strict BPB/MBE branch therefore passes the early-galaxy gate: not as a raw-likelihood triumph, but as a physically interpretable high-redshift solution with a testable host-mass shift.

A v0.3.2 source inventory

The following CSV files are shipped with v0.3.2 and are used by the figure-generation script or the claim-source lock:

- `data/raw_chi2.csv`: UVLF, stellar-mass-density, and raw chi-square values.
- `data/mapper_physical_peak.csv`: $M_{\text{UV,peak}}$ and $\log_{10} M_{\text{peak}}$.
- `data/prior_penalties.csv`: M_{peak} prior contribution, total prior penalty, and data-plus-prior diagnostic.
- `data/tau_proxy.csv`: reionization redshift and optical-depth proxy summaries.
- `data/audit_summary.csv`: display-rounding audit values and derived factors.
- `data/paperV_v0_3_numeric_locks.csv`: collected numerical locks used by the v0.3 release.
- `data/paperV_v0_3_claim_source_lock.csv`: explicit Paper-V claim/source-lock table.

All figures in this version are regenerated by

```
python scripts/make_paperV_v0_3_figures.py
```

B v0.3.2 audit lock

The numerical audit of the v0.0.0/v0.1 values produced 35/35 accepted checks after the display-rounding flags were resolved. Two known display issues remain documented rather than hidden.

First, using displayed rounded values gives

$$13.5579 - 11.2413 = 2.3166, \quad (18)$$

whereas the source calculation gives 2.3167 because the displayed value 11.2413 is itself rounded from the exact $\text{UVLF} + \rho_*$ sum. Second, the displayed data-plus-prior value 16.06 corresponds to the source value 16.0579.

Thus the v0.3.2 numerical claims are internally coherent at display precision. The key locked values are:

- $\Delta\chi^2_{\text{raw}} = +2.3167$;
- $\chi^2_{\text{data+prior,LCDM}} \simeq 17.96$;

- $\chi^2_{\text{data+prior,BPB}} = 16.0579 \simeq 16.06$;
- $M_{\text{UV,peak}}^{\text{LCDM}} = -19.1495$;
- $M_{\text{UV,peak}}^{\text{BPB}} = -19.1912$;
- $\log_{10} M_{\text{peak}}^{\text{LCDM}} = 9.9965$;
- $\log_{10} M_{\text{peak}}^{\text{BPB}} = 10.8910$;
- $\chi^2_{M_{\text{peak,LCDM}}} = 6.279$, explaining about 93% of the LCDM prior penalty.

C What v0.3.2 improves over v0.3.1

Version v0.3 added the corpus-aware source-lock layer: explicit CSV source tables, a figure-generation script, CSV-regenerated figures, and a source inventory. Version v0.3.2 is a wording hotfix: it does not alter the audited scalars, but it moves the intentionally loose host-mass regularizer and the fixed-pivot versus physical-peak interpretation into the abstract and conclusions.

References

- [1] Behroozi, P. S., Wechsler, R. H., & Conroy, C. 2013, ApJ, 770, 57.
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